U.G.C. NET Ex
COMPUTE
(Solved with Expl

1. Consider a sequence F_{00} defined as :

$F_{00}(0) = 1, F_{00}(1) = 1$

$$F_{00}(n) \frac{10 * F_{00}(n-1) + 100}{F_{00}(n-2)} \text{ for } n \geq 2$$

Then what shall be the set of values of the sequence F_{00} ?

(a) (1, 110, 1200)
(b) (1, 110, 600, 1200)
(c) (1, 2, 55, 110, 600, 1200)
(d) (1, 55, 110, 600, 1200)

Ans : (a) $F_{00}(0) = 1, F_{00}(1) = 1$

$F_{00}(n) = 10*F_{00}(n-1) + 100$

$F_{00}(2) = (10*F_{00}(1) + 100)/F_{00}(0)$

$= (10*1 + 100)/1 = 110$

$F_{00}(3) = 10*F_{00}(2) + 100)/F_{00}(1)$

$= (10*110 + 100)/1 = 1200.$

2. Match the following :

List-I		List-II	
A.	**Absurd**	**i.**	**Clearly impossible being contrary to some evident truth**
B.	**Ambiguous**	**ii.**	**Capable of more than one interpretation or meaning**
C.	**Axiom**	**iii.**	**An assertion that is accepted and used without a proof**
D.	**Conjecture**	**iv.**	**An opinion preferably based on some experience or wisdom**

Codes :

	A	B	C	D
(a)	i	ii	iii	iv
(b)	i	iii	iv	ii
(c)	ii	iii	iv	i
(d)	ii	i	iii	iv

Ans : (d)

(i) **Absurd** is an adjective, which means utterly or obviously senseless, illogical or untrue i.e., contrary to all reason or common sense or evident truth.

(ii) **Ambiguous** is an adjective, which means open to or having several possible meaning or interpretations or structure.

(iii) **Axiom** is a noun, which means a self evident truth that requires no proof or universally accepted principle or rule.

(iv) **Conjecture** is a noun, which means the formation or expression of an opinion or theory without sufficient evidence for proof i.e., an opinion base on some past experience.

3. The functions mapping R into R are defined as :

$$\mathbf{f(x) = x^3 - 4x, g(x) = \frac{1}{x^2+1} and h(x) = x^4}$$

Then find the value of the following composite functions :

$h_0g(x)$ and $h_0g_0f(x)$

(a) $(x^2 + 1)^4$ and $[(x^3 - 4x)^2 + 1]^4$

(b) $(x^2 + 1)^4$ and $[(x^3 - 4x)^2 + 1]^{-4}$

(c) $(x^2 + 1)^{-4}$ and $[(x^2 - 4x)^2 + 1]^4$

(d) $(x^2 + 1)^{-4}$ and $[x^3 - 4x)^2 + 1]^{-4}$

Ans : (d) $F(x) = x^3 - 4x$, $g(x) = \frac{1}{x^2+1}$, $h(x) = x^4$

Then $h_0g(x) = h(g(x)) = h\left(\frac{1}{x^2+1}\right)$

$$= \left(\frac{1}{(x^2)+1}\right)^4 = \frac{1}{(x^2+1)^4} = (x^2+1)^{-4}$$

Then $h_0g_0f(x) = h(g_0f(x) = h(gf(x))) = h(g(x^3 - 4x))$

$$h(g(x^3-4x)) = h\left(\frac{1}{(x^3-4x)^2+1}\right) = \left(\frac{1}{(x^3-4x)^2+1}\right)^4$$

$$= ((x^3-4x)^2+1)^{-4}$$

4. How many multiples of 6 are there between the following pairs of numbers?

0 and 100 and–6 and 34

(a) 16 and 6 (b) 17 and 6

(c) 17 and 7 (d) 16 and 7

Ans : (c)

(i) Number of multiples of 6 between 1 to $100 = \left\lfloor \frac{100}{6} \right\rfloor = 16$. Since range start from 0 and 0 is multiple of 6.

(ii) Number of multiples of 6 between 0 to $= \left\lfloor \frac{34}{6} \right\rfloor + 1(\text{for } 0) = 5 + 1 = 6.$

Since range start from -6 and -6 is also multiple of 6. So, total multiple of 6 = 6 + 1 = 7. So, 17 and 7 are multiples of 6 for both pairs respectively.

5. Consider a Hamiltonian Graph G with loops or parallel edges and with $|V(G)| = n \geq 3$. Then which of the following is true?

(a) $\deg(v) \geq \frac{n}{2}$ for each vertex v.

(b) $|E(G)| \geq \frac{1}{2}(n-1)(n-2)+2$

(c) $\deg(v)+\deg(w) \geq n$ whenever v and w are not connected by an edge.

(d) All of the above

Ans : **(d)** For hamiltonian graph 'G' with are loops or parallel edges with $|v(G)| = n \geq 3$ then:

• According to ore's theorem sum of degree of every non-adjacent pair of vertices i.e., deg (v) +d (w) must be greater than equal to n(total number of vertices).

• According to Dirac's theorem degree of every vertex must be greater than n/2.

• Since according to Dirac's theorem

$\deg(v)+\deg(w) \geq n$ ....(i)

Hence $|E| \leq |E(G-\{u,v\})| + \deg G(v) + \deg G(w)$

$\leq \binom{n-2}{2} + n$ using equation (i)

$$\leq \frac{(n-2)(n-3)}{2} + n$$

$$\leq \frac{n^2-3n-2n+6+2n}{2}$$

$$\leq \frac{n^2-3n-6}{2}$$

6. In propositional logic if (P → Q) ∧ (R → S) and (P ∧ R) are two premises such that

$$\begin{array}{c} (P \to Q) \wedge (R \to S) \\ \underline{P \vee R} \\ \underline{Y} \end{array}$$

Y is the premise :

(a) $P \vee R$ (b) $R \vee S$

(c) $Q \vee R$ (d) $Q \vee S$

Ans : **(d)** Given that premises are

$(P \to Q) \wedge (R \to S)$

$(P \vee R)$

$(P \to Q) = \sim P \vee Q$

$(R \to Q) = \sim R \vee S$

$(P \vee R)$

$Q \vee S$:- There will be resolution (rule of inference) b/w there premises to give conclusion.

~ P & P, R & R' will resolve out and then we contract are disjunction of remaining clauses to given $S \vee Q$ is correct option.

7. **ECL is the fastest of all families. High speed in ECL is possible because transistors and used in difference amplifier configuration, in which they are never driven into ....**

(a) Race condition (b) Saturation
(c) Delay (d) High impedance

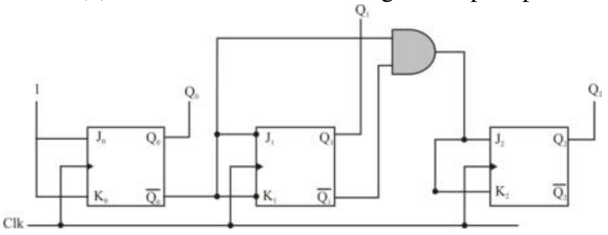
Ans : (b) ECL is fastest among the three because the transistors are used in difference amplifier configuration, in which they are never driven in saturation and there by the storage time is eliminated.

8. **A binary -bit down counter uses J-K flip-flops, FF_i with inputs J_i, K_i and outputs Q_i, i = 0, 1, 2 respectively. The minimized expression fro the input from following, is**

I. $J_0 = K_0 = 0$
II. $J_0 = K_0 = 1$
III. $J_1 = K_1 = Q_0$
IV. $J_1 = K_1 = \bar{Q}_0$
V. $J_2 = K_2 = Q_1 Q_0$
VI. $J_2 = K_2 = \bar{Q}_1 \bar{Q}_0$

(a) I, III, V (b) I, IV, VI
(c) II, III, V (d) II, IV, VI

Ans : (d) **3 bit down counter** using J-K flip-flop:



So, $J_0 = K_0 = 1$, $J_1 = K_1 = \bar{Q}_0$ and $J_2 = K_2 = \bar{Q}_1.\bar{Q}_0$

9. **Convert the octal number 0.4051 into its equivalent decimal number.**

(a) 0.5100098 (b) 0.2096
(c) 0.52 (d) 0.4192

Ans : (a) Enter the number = $(0.4051)_8$
decimal equivalent of $(0.4051)_8 = 0.8^0 + 4.8^{-1} + 0.8^{-2} + 5.8^{-3} + 1.8^{-4} = 0 + 0.5 + 0 + 0.000244140625$
$= 0.510009765625_{10} = 0.5100098$

10. **The hexadecimal equivalent of the octal number 2357 is :**

(a) 2EE (b) 2FF
(c) 4EF (d) 4FE

Ans : (c) $(2357)_8 = (010\ 011\ 101\ 111)_2$
now grouping them of 4 from right to left
0100 1110 1111 – so it will be 4EF

11. **Which of the following cannot be passed to a function in C^{++}?**

(a) Constant (b) Structure
(c) Array (d) Header file

Ans : (d) Header file cannot passes as to a function in C^{++}. While array, content and structure can be passed into a function.

12. Which one of the following is correct for overloaded functions in C++?

(a) Compiler sets up a separate function for every definition of function

(b) Compiler does not set up a separate function for every definition of function

(c) Overloaded functions cannot handle different types of objects

(d) Overloaded functions cannot have same number of arguments

Ans : **(a)** When you call an overloaded function, the compiler determines the most appropriate definition to use by comparing the arguments types you used to call the function with the parameter types specified in the definitions. The process of selecting the most appropriate overloaded function is called overload resolution.

13. Which of the following storage classes have global visibility in C/C++?

(a) Auto (b) Extern

(c) Static (d) Register

Ans : **(b)** Extern is used for declaring extern variables in C. This modifier is used will all data types like int, float, double etc. It is default storage class of all global variable as well all function.

14. Which of the following operators cannot be overloaded in C/C++?

(a) Bitwise right shift assignment

(b) Address of

(c) Indirection

(d) Structure reference

Ans : **(d)** Operators which cannot be overloaded.
conditional, . (member selection), :: (scope resolution), size of (object size information), *(member selection with, pointer to number)

15. If X is a binary number which is power of 2, then the value of X & (X – 1) is :

(a) 11.......11 (b) 00.......00

(c) 100......0 (d) 000.......1

Ans : **(b)** Let x = 2 ∧ 3 = 8 = 1000
then x – 1 = 7 = 0111
now x & (x – 1) = 0000
(here & is bitwise and) = if both in the compared position of the bits patterns are 1, the bit in the resulting bit pattern is 1, otherwise 0)

16. An attribute A of data type varchar (20) has value 'Ram' and the attribute B of data type char (20) has value 'Sita' in oracle. The attribute A has .... memory spaces and B has .... memory spaces.

(a) 20, 20 (b) 3, 20

(c) 3, 4 (d) 20, 4

Ans : **(b)** The CHAR & VARCHAR type are similar, but differ in the way they are stored and retrieved. They also differ in maximum length & in whether railing space are retained.

The CHAR and VARCHAR types are declared with a length that indicates the maximum number of characters you want to store.
The length of a CHAR column is fixed to the length that you declare when you create the table. VARCHAR (20) can also hold up to 20 characters but if character is less it will take less space so a needs 3 space (for ram)
b need 20 (for sita).

17. Integrity constraints ensure that change made to the database by authorized users do not result into loss of data consistency. Which of the following statement (s) is (are) true w.r.t. the examples of integrity constraints?

(A) An instructor Id. No. cannot be null, provided Instructor Id No. being primary key.

(B) No two citizens have same Adhar-Id

(a) (A), (B) and (C) are true
(b) (A) false, (B) and (C) are true
(c) (A) and (B) are true; (C) false
(d) (A), (B) and (C) are false

Ans : (c) Integrity means something like "be right" & consistent. The data in a database must be right and in good condition.
Domain integrity means the definition of a valid set of values for and tribute you define data type, length etc.
An instructor Id. no. cannot be null, provided instructor Id. no. being the primary key is valid.
No. two citizens how same adhar-id. is valid but, the budget of company must be zero is not a valid constraint.

18. Let M and N be two entities in an E-R diagram with simple single value attributes. R_1 and R_2 are two relationship between M and N, where as :
R_1 is one-to-many and R_2 is many-to-many.
The minimum number of tables required to represent M, N, R_1 and R_2 in the relational model are ....

(a) 4 (b) 6
(c) 7 (d) 3

Ans : (d) Two tables are created or entities M & N and third table is created for relationships R2 (many to many) for R1 no need of separate table (one to many) 1 – m relationships.
The primary key from the one side is placed as a foreign key in the 'many side'
- m – n relationships
  A new relation is created with primary key
- 1 – 1 relationship

19. Consider a schema R(MNPQ) and functional dependencies M $\rightarrow$ N, P $\rightarrow$ Q. Then the decomposition of R into R_1 (MN) and R_2(PQ) is ....

(a) Dependency preserving but not lossless join
(b) Dependency preserving and lossless join
(c) Lossless join but not dependency preserving
(d) Neither dependency preserving nor lossless join

Ans : (a) R(MNPQ) and functional dependencies M $\rightarrow$N, P $\rightarrow$ Q.
Then the decomposition of R into R1 (MN) & R2 (PQ) here clearly the dependencies are preserved but they are not lossless as both R1 & R2 cannot be formed back again as they do not have a common attribute.

20. The order of a leaf in a B+ tree is the maximum number of children it can have. Suppose that block size is 1 kilobyte, the child pointer takes 7 bytes long and search field value takes 14 bytes long. The order of the leaf node is ....

(a) 16 (b) 63
(c) 64 (d) 65

Ans : (a) Data pointer = child pointer + search field value = 14 + 7 = 21B
So max number of children it can have = 1 KB/21B = 1024/21 = 48.76 = 45 (approx)
All other option are greater than 48.

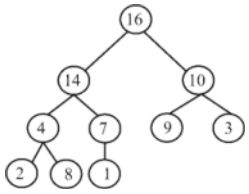
21. Which of the following is true for computation time in insertion, deletion and finding maximum and minimum element in a sorted array?

(a) Insertion – 0(1), Deletion – 0(1), Maximum – 0(1), Minimum – 0(1)
(b) Insertion – 0(1), Deletion – 0(1), Maximum – 0(n), Minimum – 0(n)
(c) Insertion – 0(n), Deletion – 0(n), Maximum – 0(1), Minimum – 0(1)
(d) Insertion – 0(n), Deletion – 0(n), Maximum – 0(n), Minimum – 0(n)

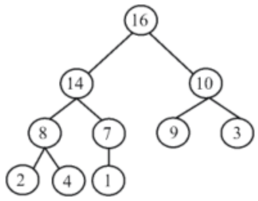
Ans : (c) Array is sorted, so in worst case(last item insert) Insertion = O(n), in worst case, last item delete deletion O(n) maximum and minimum is always at end.
maximum $\rightarrow$O(1), minimum $\rightarrow$ O(1)

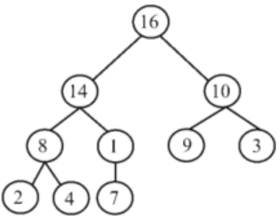
22. Which of the following is a valid heap?

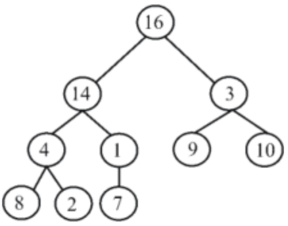
(A)



(B)

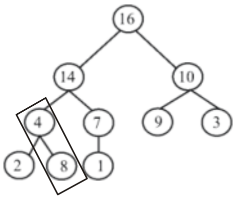


(C) 

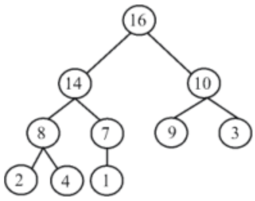
(D) 

(a) A (b) B
(c) C (d) D

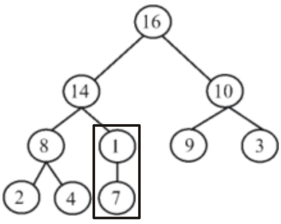
Ans : (b)

(A) 

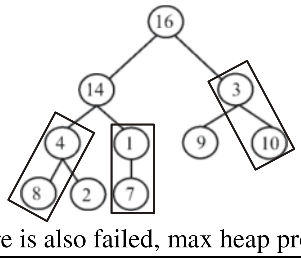
Max heap property failed because parent ≥ child

(B) 

valid max heap because every parent ≥ child

(C) 

Here is also failed, max heap property.

(D) 

Here is also failed, max heap property.

23. If h is chosen from a universal collection of hash functions and is used to hash n keys into a table of size m, where n ≤ m, the expected number of collisions involving a particular key x is less than ....

(a) 1 (b) 1/n
(c) 1/m (d) n/m

Ans : **(b)** A, D, C are invalid as there are child > parent case A. 8 > 4, C : 7 > 1, D : 10 > 3 only B is valid heap parent > children

24. Which of the following statements is false?

(A) Optimal binary search tree construction can be performed efficiently using dynamic programming.

(B) Breadth-first search cannot be used to find connected components of a graph.

(C) Given the prefix and postfix walks of a binary tree, the tree cannot be re-constructed uniquely.

(D) Depth-first-search can be used to find the connected components of a graph.

(a) A (b) B
(c) C (d) D

Ans : **(a)** Thought it is not mentioned in the option that a given. It is a scenario of universal hashing.

25. Which of the following statements is false?

(A) Optimal binary search tree construction can be performed efficiently using dynamic programming

(B) Breadth-first search cannot be used to find connected components of a graph

(C) Given the prefix and postfix walks of a binary tree, the tree cannot be re-constructed uniquely

(D) Depth-first-search can be used to find the connected components of a graph

(a) A (b) B
(c) C (d) D

Ans : **(b)** BFS and DFS both algorithm are used to find connected component. In BFS, a search that begins at some particular 'N' will find the entire connected component V before returning.

26. Match the following layers and Protocols for a user browsing with SSL :

A.	**Application of layer**	**i.**	**TCP**
B.	**Transport layer**	**ii.**	**IP**
C.	**Network layer**	**iii.**	**PPP**
D.	**Datalink layer**	**iv.**	**HTTP**

Codes :

	A	**B**	**C**	**D**
(a)	iv	i	ii	iii
(b)	iii	ii	i	iv
(c)	ii	iii	iv	i
(d)	iii	i	iv	ii

Ans : **(a)**
(a) application of layer uses HTTP (Hyper text transfer protocol) (iv)
(b) Transport layer uses TCP (Transfer control protocol) (i)
(c) Network layer uses IP (internet protocol)
(d) Datalink layer (iii) ppp

27. The maximum size of the data the application layer can pass on to the TCP layer below is ......

(a) 2^{16} bytes
(b) 2^{16} bytes + TCP header length
(c) 2^{16} bytes – TCP header length
(d) 2^{15} bytes

Ans : (*) There is no maximum limit of data that application layer can send to TCP, so all options are wrong. The default TCP maximum segment size is 536 answer should be any size.

28. A packet whose destination is outside the local TCP/IP network segment is sent to.........
(a) File server (b) DNS server
(c) DHCP server (d) Default gateway

Ans : (d) Default gateway serves an access point or IP router that a networked uses to send information to a computer in another network or the internet. Default simply means that this gateway is used by default, unless an application specifies another gateway.

29. Distance vector routing algorithm is a dynamic algorithm. The routing tables in distance vector routing algorithm are updated ....
(a) automatically
(b) by server
(c) by exchanging information with neighbor nodes
(d) with back up database

Ans : (c) A distance vector routing protocol requires that a router inform its neighbors about its topology changes periodically.

30. In link state routing algorithm after construction of link state packets, new routes are computed using :
(a) DES algorithm
(b) Dijkstra's algorithm
(c) RSA algorithm
(d) Packets

Ans : (b) Link state protocols, sometimes called shortest path first are built around a well known algorithm from graph theory Dijkstra's shortest path algorithm, which is a graph search algorithm that solves the single-source shortest path problem for a graph with non-negative edges path costs, producing a shortest path tree.

31. Which of the following strings would match the regular expression : p+[3 – 5]*[xyz]?

I.	**p443y**	**II.**	**p6y**
III.	**3xyz**	**IV.**	**p35z**
V.	**p353535x**	**VI.**	**ppp5**

(a) I, III and VI only (b) IV, V and VI only
(c) II, IV and V only (d) I, IV and V only

Ans : (d) P + [3 – 5]* [xyz] = p + (3 + 4 + 5)*(x+y+z)
Since P has to be present as the first character of each string
(I) p4434 → matched
(II) p64 → not matched since 6 cannot used

(III) 3xyz → not matched string must start with p
(IV) p35z → matched
(V) p353535X → matched
(VI) ppp5 → not matched strings must end with x, or y, or z

32. Consider the following assembly language instructions :
mov al, 15
mov ah, 15
xor al, al
mov cl, 3
shr ax, cl
add al, 90H
adc ah, 0
What is the value in ax register after execution of above instructions?

(a) 0270H (b) 01070H
(c) 01E0H (d) 0370H

Ans : (a) We know that Ax is a register containing al, ah.
MOV al, 15 → AX = $\underline{al}\,\underline{ah}$ 0F H;]
MOV ah, 15 → AX = 0F0FH;
XOR al, al → AX = 0F00H;
MOV cl, 3 → AX = 0F00H; cl = 03
SHR ax, c1 → AX = 01E0H; shift right 3 times
Add al, q ∘ h → AX = 0170H;
ADC ah, 0 → AX = 0270H; cy = 1 (carry)
ADC = add carry
So output will be AX = 0270H

33. Consider the following statements related to compiler construction :
I. Lexical Analysis is specified by context-free grammars and implemented by pushdown automata.
II. Syntax analysis is specified by regular expressions and implemented by finite-state machine.
Which of the above statement (s) is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

Ans : (d) Both statements are wrong because lexical analyzer uses finite automata so it uses regular grammar, whose expression will be for example latter (letter + digit)*, where as syntax tree uses context free grammar which uses pda.

34. The contents of register (BL) and register (AL) of 8085 microprocessor are 49H and 3AH respectively. The contents of Al, the status of carry flag (CF) and sign flag (SF) after executing 'SUB AL, BL' assembly language instruction, are

(a) AL = 0FH; CF = 1; SF = 1
(b) AL = F0H; CF = 0; SF = 0
(c) AL = F1H; CF = 1; SF = 1
(d) AL = 1FH; CF = 1; SF = 1

Ans : (d)
(A) ⇒ 3AH ⇒ 00111010
(B) ⇒ 49H ⇒ <u>01001001</u>
11110001
(A) ⇒ F1H, (CY) ⇒ 1, (S) ⇒ 1
The carry flag is set since the first operand is less than the second operand. Since the result produce the negative result sign flag is set.

35. Which of the following statements (s) regarding a linker software is/are true?
I. A function of a linker is to combine several object modules into a single load module.
II. A function of a linker is to replace absolute references in an object module by symbolic references to locations in other modules

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

Ans : (a) It is saying that linker replace absolute references in an object which is wrong linker always generate replaceable address.

36. There are three processes P1, P2 and P3 sharing a semaphore for synchronizing a variable. Initial value of semaphore is one. Assume that negative value of semaphore tells us how many processes are waiting in queue. Processes access the semaphore in following order :

(A) P2 needs to access
(B) P1 needs to access
(C) P3 needs to access
(D) P2 exits critical section
(E) P1 exits critical section

The final value of semaphore will be :

(a) 0 (b) 1
(c) –1 (d) –2

Ans : (a) Initial value of semaphores S = 1
(1) P2 needs to access decrease it to 0 (no one is waiting)
(2) P1 needs to access decreases it to –1 (one process in waiting)
(3) P3 needs to access decreases it to – 2 (2 process are waiting
(4) P2 exits critical section increases it to –1 (one process is waiting)
(5) P1 exits critical section increases it to –0 (no process is waiting)

37. In a paging system, it takes 30 ns to search translation Look-a-side buffer (TLB) and 90 ns to access the main memory. If the TLB hit ratio is 70%, the effective memory access time is :

(a) 48ns (b) 147ns
(c) 120ns (d) 84ns

Ans : (b) 70% TLB hit ratio 30 ns to search TLB
(0.7 * 30) in hit case : to find the page
0.3*(90 +30) in miss case : to find the page
90 ns to access the data from main memory
So total time : (0.7 * 30) + (0.3 * (90 + 30)) + 90 = 147

38. Match the following w.r.t. Input/Output management :

List-I		List-II	
A.	**Device controller**	**i.**	**Extracts information from the controller register and store it in data buffer**
B.	**Device driver**	**ii.**	**I/O scheduling**
C.	**Interrupt handler**	**iii.**	**Performs data transfer**
D.	**Kernel I/O subsystem**	**iv.**	**Processing of I/O request**

Codes :

	A	**B**	**C**	**D**
(a)	iii	iv	i	ii
(b)	ii	i	iv	iii
(c)	iv	i	ii	iii
(d)	i	iii	iv	ii

Ans : (d)

(i) **Interrupt handler** extracts information from the controller register and store it in data buffer.

(ii) **Device driver** process the I/O request.

(iii) **Device controller** performs data transfer.

(iv) **Kernal I/O subsystem** perform I/O scheduling.

39. Which of the following scheduling algorithms may cause starvation?

A. First-come-first-served

B. Round Robin

C. Priority

D. Shortest process next

E. Shortest remaining time first

(a) A, C and E (b) C, D and E

(c) B, D, and E (d) B, C and D

Ans : (b)

(1) First come – first served – No starvation

(2) Round robin – No starvation

(3) Priority – starvation if higher priority process called again & again lower priority starves

(4) Shortest process next = saturation possible

(6) Shortest remaining time first = starvation possible

40. Distributed operating system consist of :

(a) Loosely coupled O.S. software on a loosely coupled hardware

(b) Loosely coupled O.S. Software on a tightly coupled hardware

(c) Tightly coupled O.S. Software on a loosely coupled hardware

(d) Tightly coupled O.S. software on a tightly coupled hardware

Ans : (c) A distributed operating system is usually defined as running on more loosely coupled hardware. There is no shared memory, but provides a single memory by tightly coupled software.

41. Software Engineering is an engineering discipline that is concerned with :

(a) how computer systems work
(b) theories and methods that underlie computers and software systems
(c) all aspects of software production
(d) all aspects of computer-based system development, including hardware, software and process engineering

Ans : (c) Software engineering is an engineering discipline that is concerned with all aspects of software production.

42. Which of the following is not one of three software product aspects addressed by Mccall's software quality factors?

(a) Ability to undergo change
(b) Adaptability to new environments
(c) Operational characteristics
(d) Production costs and scheduling

Ans : (d) Mccalls's software quality factors.
(1) ability to undergo change
(2) Adaptability to new environments
(3) Operational characteristics
but production costs & scheduling is not the aspect of quality factor.

43. Which of the following statement(s) is/are true with respect to software architecture?

S1 : Coupling is a measure of how well the grouped together in a module belong together logically.

S2 : Cohesion is a measure of the degree of interaction between software modules

S3 : If coupling is low and cohesion is high then it is easier to change one module without affecting others.

(a) Only S1 and S2
(b) Only S3
(c) All of S1, S2 and S3
(d) Only S1

Ans : (b) S3 is true because, If coupling is low and cohesion is high then it is easier to change one module without affecting others.

44. The prototyping model of software development is :

(a) a reasonable approach when requirements are well-defined
(b) a useful approach when a customer cannot define requirements clearly
(c) the best approach to use for projects with large development teams
(d) a risky model that rarely produces a meaningful product

Ans : (b) Prototype model is a useful approach when a customer cannot define requirements clearly.

45. A software design pattern used to enhance the functionality of an object at run-time is :

(a) Adapter (b) Decorator
(c) Delegatio (d) Proxy

Ans : (b) In object oriented programming, the decorate or pattern (also known as wrapper, an alternative naming shared with adapter pattern) is a design pattern that allows behavior to be added to an in dividable object, either statically or dynamically, without effecting the behavior of other object from the same class.

46. Match the following :

List-I		List-II	
A.	**Affiliate Marketing**	**i.**	**Vendors ask partners to place logos on partner's site. If customers click, come to vendors and buy**
B.	**Viral Marketing**	**ii.**	**Spread your brand on the net by word-of-mouth. Receivers will send your information to their friends**
C.	**Group Purchasing**	**iii.**	**Aggregating the demands of small buyers to get a large volume. Then negotiate a price.**
D.	**Bartering Online**	**iv.**	**Exchanging surplus products and services with the process administered completely online by an intermediary. Company receives "points" for its contribution.**

Codes :

	A	B	C	D
(a)	i	ii	iii	iv
(b)	i	iii	ii	iv
(c)	iii	ii	iv	i
(d)	ii	iii	i	iv

Ans : (a) Affiliate Marketing: Vendors ask partners to place logos on partner's site. If customers click, come to vendors and buy.

Viral Marketing: Spread your brand on the net by word-of-mouth. Receivers will send your information to their friends.

Group Purchasing: Aggregating the demands of small buyers to get a large volume. Then negotiate a price.
Bartering Online: Exchanging surplus products and services with the process administered completely online by an intermediary. Company receives "points" for its contribution.

47. .... refers loosely to the process of set automatically analyzing large databases find useful patterns

(a) Datamining (b) Data warehousing
(c) DBMS (d) Data mirroring

Ans : (a) The practice of examining large pre-existing databases in order to generate new information is data mining. It refers loosely to the process of semi-automatically analyzing large database to find useful patterns.

48. Which of the following is/are true w.r.t. application of mobile computing?

(A) Traveling of salesman
(B) Location awareness services

(a) (A) true; (B) false
(b) Both (A) and (B) are true
(c) Both (A) and (B) are false
(d) (A) false; (B) true

Ans : (b) With respect to application of mobile computing
(1) Traveling of salesman – true
(2) Location awareness service – true

49. In 3G network, W-CDMA is also known as UMTS. The minimum spectrum allocation required for W-CDMA is ....

(a) 2 MHz (b) 20 KHz
(c) 5 KHz (d) 5 MHz

Ans : (d) In 3G network, W-CDMA is also known as UMTS. The minimum spectrum allocation required for W-CDMA is 5MHz

50. Which of the following statements is/are true w.r.t. Enterprise Resource planning (ERP)?

(A) ERP automates and integrates majority of business processes
(B) ERP provides access to information in a Real Time Environment.
(C) ERP is inexpensive to implement.

(a) (A), (B) and (C) are false
(b) (A) and (B) false; (C) true
(c) (A) and (B) true; (C) false
(d) (A) true; (B) and (C) are false

Ans : (c)
(1) ERP automates and integrates majority of business – true
(2) ERP provides access to information in a real time environment – true
(3) ERP is inexpensive to implement – false